

What Does This Test Support?

Unit 4 · Topic 4.10 · TODAY'S PROBLEM

Suppose 4,000 active students chose Focus mode and 3,600 chose Guided mode on an online practice service this term. Independent SRSs of 100 nonoverlapping students completed the same set; summaries are shown. Before sampling, the service asked whether $\mu_F > \mu_G$ and set $\alpha = 0.05$. Students chose their modes, and only a difference of at least 3 minutes is practically important.

Nia: “Since $p \approx 0.0263 < 0.05$, there is evidence that the Focus population mean is higher. Therefore Guided mode causes every student to finish at least 3 minutes faster; switch everyone.”

Mateo: “The observed gap is only 1.2 minutes and students selected their modes, so neither a 3-minute effect nor causation is established. Therefore retain H_0 ; there is no evidence of a population mean difference.”

Practice time summaries (minutes)

Mode	N	n	$\bar{x}$	s
Focus	4,000	100	42.8	4.2
Guided	3,600	100	41.6	4.5

FIRST TAKE (5 min, on your sheet)

Which clauses in Nia’s and Mateo’s reports look defensible? Cite one number and one design feature.

- DOK 1** (a) Define μ_F and μ_G , then state H_0 and H_a in Focus-minus-Guided order.
- DOK 2** (b) Check the two-sample t conditions. Compute the difference, SE , t , technology degrees of freedom, and one-sided p -value; interpret the p -value.
- DOK 3** (c) Evaluate both reports and write one warranted conclusion. Coordinate p versus α , the 1.2-minute gap versus the 3-minute benchmark, means versus individuals, SRS population scope, and self-selection. Explain the decision risk of retaining either report unchanged.

Video + follow-along: [u7_lesson9_live.html](https://www.khanacademy.org/a/u7_lesson9_live.html) (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

